Instructions to build Minimum Viable Data

Goal
I want to test if AI can product the data I want. My main goal is the do the ‘Quality Check the dat provided by AI’

Let’s call this release the ‘MVD -Minimum viable data’. Ultimately, I want to create a product (a data/decision intelligence tool) that helps surface insights from a company’s data. But for now, we don’t need to consider the user experience. I just want to see the quality of the data AI can produce.

Description
Test if AI can produce the data I want. The data the AI will ingest includes 1) user/company information (collected from onboarding) 2) the data from the company (provided by the users by csv, in future it’ll be API/integrations) 3) publicly available information (e.g. publicly available benchmark data).

The processed data is to be saved to a database. And rendered visually for the user.

The key actions for the user are scroll (scroll through different KPI overviews) and explore (user can explore in depth analysis and synthesis for a kpi) and receive weekly brief (get a weekly update on what’s important)

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Other tasks

- ☐ Collect input data from customer
- ☐ Show what the final UI should look like.
- ☐ Should what the key interactions will be. How system should respond.
- ☐ Create agent instructions to process data
- ☐ Create potential critical path/overall workflow to create the data

Define key use cases

UC	Who	Problem	Current solutions	Purpose/ Intended outcome	Frequency	Proposed feature	Data required	Logic/ Processing required
1	Admin/ Exec/CSM	Customer has goals, KPIs. Some they know, some they don't know. All of our customers (in our segment) are trying to get over the next 'hump', in	Searching GPT for strategic advice Research for benchmark companies. Board, advisors.	Having a better idea of what to focus on, helps prioritise time, energy, resources	Setting goals: Monthly Reporting on goal performanc e: Weekly, fortnightly,	KPI recommen dation via Strategy Consultant Agent	User context table. E.g. User role. User KPIs, upcoming 'what's important' Org context	Strategy Consultant Agent to determine what KPIs are relevant (leading), to achieve their goals

		the startup, business cycle. *Problem: What should we focus on? What goals, KPIs? Are we aligned?*	Asking professional peers		monthly		table. E.g. Org goal, KPIs, business model, product category, org stage, product stage	(lagging).
2	CSM	Pulling data from tools takes time. Pulling data via engineers, data analysis is bottle neck. *Problem: Difficult, slow to pull*	Saving report from tools Asking devs for reports.	Instant access and ability to understand the data, with minimal friction.	Daily	Ingest - CSV + Integration s via Data Engineer Agent	API for Zoho Desk, CRM API for custom data	Data Engineer Agent helps with data pipelines, flow, normalisation etc.
3	CSM	Problem: Cleaning up the raw data, data hygiene.	User manually opens spreadsheets, cleans up columns, fields etc.	Makes sure data is ready to be analysed, to give clean, accurate results.	Weekly	Data hygiene via Data Analyst Agent	All	Data Analyst Agent. E.g. identifies data quality issues, optimises database structure, designs data strategies
4	CSM	Problem: Making sure data is being calculated, anlaysed properly.	User manually creates formulas, calcs. Open to risk, wrong calcs.	Analyse the data, summarise the data.	Daily.	SQL calcs, analysis via Business	All	Business intellgenent Agent. E.g. validates

						Intelligence Agent		formulas and calcs. Thinks through SQL.
5	CSM/Exec	Making sense of the data. What does it mean for my role, the business? *Problem: Exec has no 'so what' from Manager* - they bring raw(ish) data to exec updates.	Saving report, uploading to LLM. Asking LLM some questions (that aren't the right questions). Not sure if answer is correct. Loss of context.	Faster, better quality 'so what' from the data. Leads toFaster and better decisions, by having better sense of what needs to be done.	Weekly	Signal Synthesis (each signal) Weekly brief (prioritised top 5 signals this week) All via Management Consultant Agent,	Data tables user has provided, E.g. Zoho CRM, Custom tables User context table Org context table	Management Consultant agent. Synthesises data to provide McKinsey style synthesis.
6	Exec	Exec wants a quick exec explore of key metrics, e.g. at beginning of the week, or before meetings. *Problem: Takes days, weeks, for team to collect, analyse data for exec to have a quick 'get upto speed' summary*	Managers/CSM produces reports. Self-decides what's important to discuss with exec. Shows exec	For exec to quickly identify priority parts of business to focus on. E.g. risks, issues bubbling up, fires to address (what to slow down) But also opportunities (what to speed	Daily Weekly	Signal Cards Scrolling		Chief of Staff Agent - Decides what's important and relevant to CEO, users.

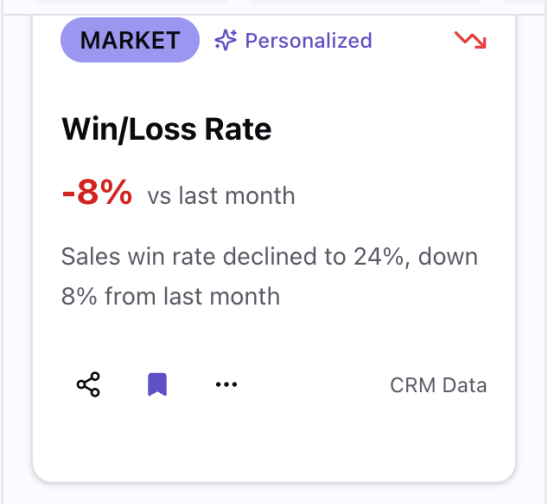
				up)				
7	Exec	Exec wants a quickly weekly overview, at the beginning of the week. *Problem: I have limited focus, where should I allocate my attention so that it has greatest impact on our goals?*	Exec has a business coach. Exec has to manually decide, based on gut check. Exec does not smartly focus attention, focuses on the wrong things Exec spends time considering strategy, roadmap, intentionally finds information related to goals.	Increase time and focus exec spends on issues that matter - e.g. help achieve goals. Minimise time/restrictions on issues that don't impact the goals.	Weekly Mid-week	Weekly brief	Signals-Analysis Signals-Synthesis	Chief of Staff agent, in collaboration with Management Consultant Agent. Helps CEO focus on priority matters.
8	CSM	Problems come up. Need explore, verify, understand actual issue, gather more context. *Problem: I've noticed something. need to understand more about what's happening here*	Exploring reports. Talk to others. Being behind the 8 ball in terms of identifying problems, opps.	People will want to validate, verify, develop a better understanding of an issue. Esp. when there's low trust with AI.	Daily			
9	CSM/Exec	Users don't have	User another LLM to	Verify what user	Onboarding	Human-AI	User input	Feedback from

		much trust with AI outputs. They have experienced hallucinations, wrong answers etc. in their personal and work interactions with LLM AIs Problem: AI outputs aren't often trusted.	verify Verify against their own data. Check with colleagues,.	is seeing, understanding. Make sure it's 'true'.	When data is ingested When data is processed	Trust Loop	data to confirm. E.g. when strategic agent identifies KPIs, user to confirm which are priority. E.g. When Management Consultant provides synth, user to provides feedback on whether it was useful	user to update the agent. But also the data we store.

Show UI + Explain data requirements

Note: Screens are for lo-fi prototypes. Final output can differ.

KPIs user has chosen, sufficient data

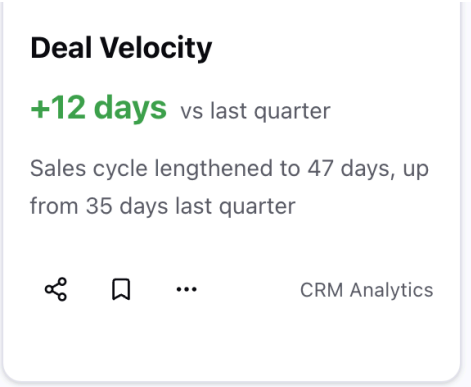
View	Screen	Description	Data requirements
1		Name: Signal-Overview User goal: Quick scroll of KPIs. 'Metrics in my pocket'. Get upto speed in beginning of week, or before meetings.	KPI - E.g. win/loss Current value - E.g. 24% Change - E.g. Down 8% from last week 1 line summary - E.g. Sales win rate declined to 24%, down from 8% last week.

2	**Implications** Declining win rate impacts revenue targets and may indicate need for pricing adjustments or product enhancements to remain competitive. **Next Steps** 1. Conduct win/loss analysis interviews 2. Review pricing strategy 3. Address top feature gaps 4. Improve sales qualification process	Name: Signal-So what User goal (exec): Exec can understand, or at least have some ideas, about what they should do with this new information User goal (manager): User can start to understand how to develop a 'so what', so that they can perform their own so what, and discuss with exec	1. 1 line exec summary — What the data shows E.g. Win rate has decreased significantly due to increased competition and longer sales cycles. Enterprise deals facing more scrutiny and budget constraints. 2. Takeaway Breakdown — Was change directionally good or bad? Was change expected or unexpected? 3. Benchmark Comparison — How it compares to norms (and specifics about who informed those norms, e.g. product category, business stage, company name) 4. Root Cause Analysis — Why it happened? Deeper drivers. 4.1 Root cause analysis rational - Chain of though, steps of the al logic of how it came to this conclusion 5. Relationship with other metrics - ● Signal relationships - Other KPIs that have a high probability of being correlated, including cross-source. ● Causal chain analysis - maps signal relationships: Identifies which signals influence others, Traces sequences of cause-and-effect (e.g., "churn up → NPS down → revenue at risk") ● Detects leading indicators (signals that typically precede others, summarises causal networks for complex multi-source scenarios 6. Implications on User/Org KPIs (Predictive signals): Potential Impact on user KPIs, if change continues. Impact on Business KPIs if change continues

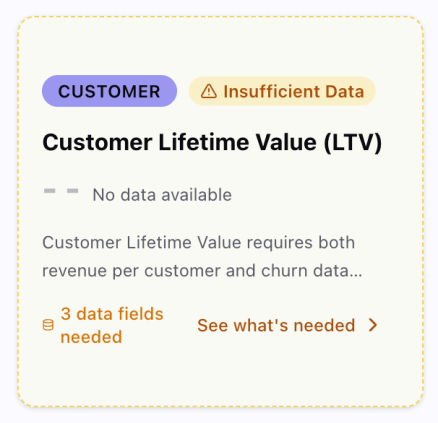
			7.
3.	process **Data Sources** Salesforce CRM Opportunity and deal closure data Gong.io Sales call analysis and competitor mentions	Name: Signal-What we found User goal: Check the ‘provenance’ of the KPI. Verify it matches data from source, what other user’s see in the source. ‘Can I trust this data, this analysis?’	Data origin tool: What was dats source? E.g. Zoho CRM How was this KPI calculated? Show calcs, including fields for normalised and calculations. User needs to be able to understand and agree. Sample data: Show a sample of the data that was used, including column name and data field
4	``` YOUR SIGNALS THIS WEEK Pipeline Value $1.24M → flat stable Win Rate 38% ↑ +6pts improving Closed Revenue $142K ↓ -$18K watch Avg Sales Cycle 47 days → flat stable Ticket Volume 84 ↑ +12 watch Avg Resolution 6.2 hrs ↓ -1.1 hrs improving Lead Conversion 14% → flat stable ``` ...	Name: Weekly Brief User goal: Given 1min at beginning of the week, what does a CEO need to care about?	Section 1: Signals this week **Each row contains:** - KPI name - Current value (absolute) - Delta from last period (arrow + number) - Status badge: `improving` / `stable` / `watch` / `alert` **Status badge rules:** | Badge | Condition | |---|---| | `improving` | Delta > threshold in positive direction | | `stable` | Delta within ±threshold | | `watch` | Delta > threshold in negative direction, not yet critical |

				`alert`	Delta > 2x threshold in negative direction, or threshold crossed | Section 2: Top 3 issues this week The top 3 most important signal this week gets one paragraph of interpretation each. **Rules:** - Maximum 5 sentences - Must reference the trend (not just this week's number) - Must give one forward-looking statement ("watch whether X") - Must not recommend an action — that's the exec's job THIS WEEK'S FOCUS Ticket Volume – 84 tickets (+12 from last week) Volume has been rising for three consecutive weeks: 60 → 72 → 84. At this rate it will cross 100 tickets per week by mid-April. The increase correlates with the onboarding of three new pharmacy groups in late February. Resolution time has not yet been affected (6.2 hrs, within target). Watch whether resolution time follows volume up in coming weeks.

KPIs user has not chosen, but relevant (as recommended by strategy agent, sufficient data)

View	Screen	Description	Data requirements
5		Name: Signal-Overview (recommended) User goal: Learn and understand what KPIs they should be considering. User can filter 'recommended signals' on and off	These are exactly the same as all other signal cards, except it has a tag for 'recommended', so that user can only show signals they've explicitly chosen, or see recommended signals as well.

KPIs user has chosen, or we have recommended, but have, insufficient data

View	Screen	Description	Data requirements
6		Name: Signal-Insufficient data User goal: Know what KPIs are important AND what additional data they need to calculate and analyze the signal.,	An agent needs to analyse the KPI, and current data, to determine what data is missing. Then provide suggestions of what data they need to sufficiently calculate/analyse the signal. Explain why that data is important to collect.

Provide Input data to AI models

- Exec user's file (see attached)
- Manager user's file (see attached)
- Zoho CRM data
- Zoho Desk data
- User's custom data file

Run AI Agents instructions to process data

Strategy Consultant
Expertise: Metric prioritization (for a particular function/role), business strategy, decision frameworks
How they help:

- Recommends the metrics a user should check weekly (e.g. CEO, CSM)
- Connects each metric to a specific business decision
- Considers industry, stage, team size, and priorities
- Filters out vanity metrics in favor of actionable ones
- Thinks in terms of what drives business outcomes

Data Engineer agent
Expertise: Data pipelines, ETL, schema mapping, data contracts, integration
How they help:

- Checks column mappings are correct
- Validates data types and parsing logic
- Identifies edge cases in source data
- Designs robust data contracts
- Ensures schema validation
- Thinks in terms of data flow and transformation

Data analyst agent
Expertise: Data modeling, SQL optimization, analytics, metrics, reporting

How they help

- Optimizes database structure
- Writes efficient SQL queries
- Identifies data quality issues
- Suggests meaningful metrics
- Designs aggregation strategies

Business intelligence agent
Expertise: Business intelligence, metric calculations, SQL, data validation
How they help:

- Validates formulas and calculations
- Thinks through SQL logic and edge cases
- Identifies missing filters or wrong assumptions
- Checks date range handling
- Ensures nulls and duplicates are handled correctly
- Always asks to see raw data before confirming numbers

Management Consultant Agent
Expertise: SMB operations, metric prioritization, business strategy, decision frameworks
How they help:

- Recommends the 5 metrics a CEO should check weekly
- Connects each metric to a specific business decision
- Considers industry, stage, team size, and priorities
- Filters out vanity metrics in favor of actionable ones
- Thinks in terms of what drives business outcomes

Quality Check the data provided by AI

- How does it look on a signal card - overview?
- How does it look on a signal card - expanded?
- Are signal calcs, changes correct? Verified?
- Is signal analysis useful, relevant, believable, able to be quality checked?
- How does a signal card - insufficient data, look?

Draft Tech architecture

Goal: For now, I just want to see if AI can produce the data I want. In future, we will need to consider UX, UI, scalability, AI costs etc. I want to test the AI data, with the least amount of complexity.

Front end: Vue & Nuxt
Backend: NodeJS
Database: ?
AI: Gemini or Open AI or Claude

Draft Critical Path / Overall workflow to create data

- User 1 (exec user) provides their information via onboarding. Saved to ‘Exec user file’.
- User 2 (manager user) provides their information via onboarding. Saved to ‘CSM user file’
- User 2/Admin connects their data. Data files saved to system.
- AI agents work, in parallel, or in succession, in to save data to database, enough to produce data required in main use cases and key user views.
- At certain steps, e.g. after data is ingested for the first time for a data shape, there ia a Human in the loop. E.g. Where a user verifies normalised columns names. So that user can trust the data hygiene, normalisation.
- Data is rendered visually to create signal cards, so that user can scroll (in scroll view) and deep dive (when they explore a signal card)

- Human in the loop - User can provide feedback on usefulness of the processed data, i.e. data in the signal cards, manually via a call.